

suiting to this system.

According to the ST25DV64K datasheet published by STMicroelectronics, the IC provides 64-kbit (8kB) of user EEPROM memory, which can be divided into up to four password-protected areas. It supports ISO/IEC 15693 and NFC Forum Type 5 standards, ensuring compatibility with most NFC-enabled smartphones.

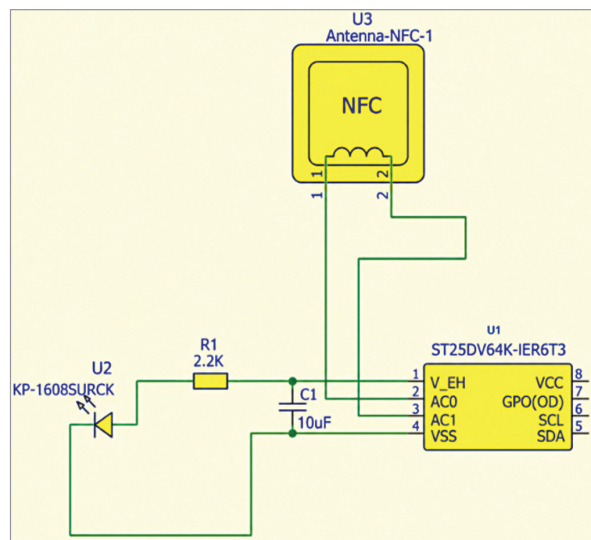


Fig. 2: Circuit diagram of NFC business card

The IC also features an I²C interface supporting data rates of up to 1MHz, a 256-byte FTM mailbox for bidirectional communication between the RF and I²C interfaces, and an energy harvesting output (V_{EH}) for powering low-power external circuits. The ST25DV64K family is available in multiple package options. The compact TSSOP-8 package was selected for

this system because it is well suited to both hand soldering and small-batch SMD assembly.

Circuit and working of the NFC business card

Fig. 2 shows the circuit diagram of the

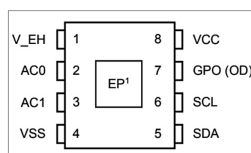


Fig. 3: PCB footprint of the ST25DV64K-IER6T3 IC

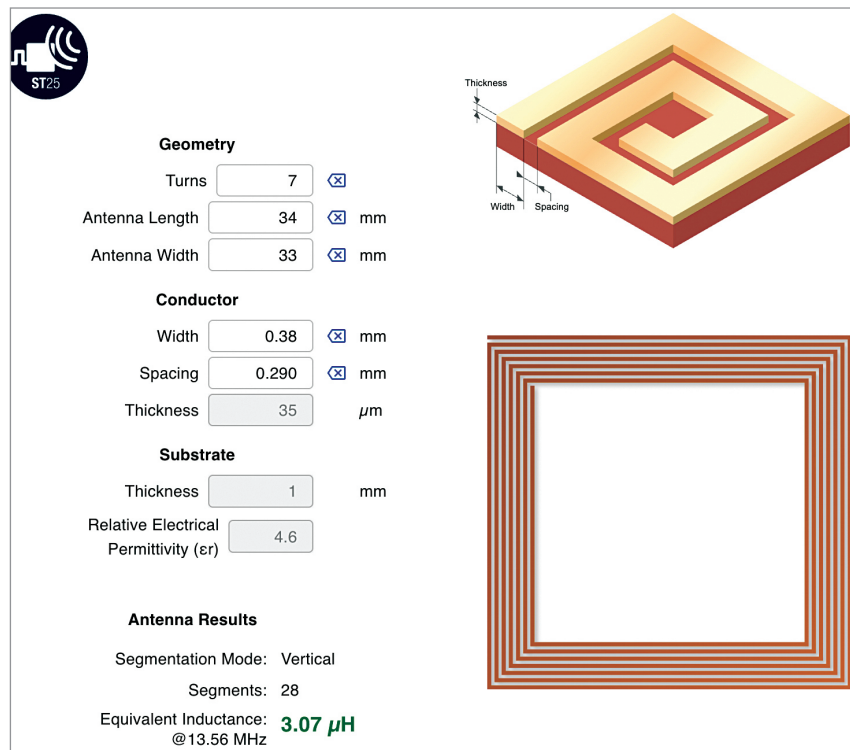


Fig. 4: NFC antenna design tool

NFC business card. The design is based on the ST25DV64K-IER6T3 (U1) dynamic NFC/RFID tag IC, which is connected to the NFC antenna (U3) through the AC0 and AC1 pins. The antenna forms a resonant circuit that receives the 13.56MHz RF signal emitted by an NFC-enabled smartphone when it is brought close to the card.

The captured RF energy powers the ST25DV64K-IER6T3, allowing the IC to communicate wirelessly with the smartphone and transfer stored data such as contact details, website URLs, or digital business cards. In addition to data communication, the IC provides an energy-harvesting output through the V_{EH} pin.

In this system, the harvested energy drives the KP-1608SURCK hyper-red LED (U2) through a 2.2kΩ current-limiting resistor (R1). When an NFC-enabled smartphone enters the operating range, it energises the NFC tag, activating the V_{EH} output and illuminating the LED. The illuminated LED provides a visual indication that the card is being read. A 10μF capacitor (C1), connected between V_{EH} and VSS (GND), smooths and stabilises the harvested voltage during RF field variations. The VCC pin may be left unconnected or used when powering the IC externally for programming, while the GPO pin is available for optional interrupt or status functions.

This circuit demonstrates the wireless data transfer and energy-harvesting capabilities of the ST25DV64K-IER6T3. Under typical conditions, the IC can harvest approximately 2mW-3mW of power in weak RF fields and up to 10mW-20mW in strong RF fields. This is sufficient to illuminate the LED and power other low-power external circuits. Fig. 3 shows the PCB footprint of the ST25DV64K-IER6T3 IC.

NFC antenna design

The antenna is the most critical part